# Kabelo Rabalao

Software Engineering Student | Game & Systems Development
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## Profile

Software engineering student with a focus on backend development - building APIs, designing data models, and writing systems that are reliable and easy to maintain. Comfortable working close to the stack, with a preference for understanding how things work rather than just making them work. Looking to apply and deepen that foundation in a team environment that values clean, thoughtful engineering.

## Education

WeThinkCode_ - Occupational Certificate in Software Engineering(NQF 6)
***Feb 2025 – Present(Expected May 2026)***

- Data Structures & Algorithms
- Object-Oriented Programming
- Networking & Socket Programming
- Database Design
- Software Development Lifecycle

Crystal Park High – National Senior Certificate(NQF 4)
***Dec 2022***
Subjects:

- English Home Language
- Afrikaans First Additional Language
- Mathematics
- Life Sciences
- Physical Sciences
- Life Orientation
- Accounting

## Technical Skills

**Languages**
Java • Python • C# • JavaScript
**Databases & Tools**
SQLite • Docker • Linux (CLI)
**Version Control**
Git • GitHub • GitLab
**Core Concepts**
Multithreading • Socket Programming • API Integration • JSON Serialization • Command-Line Interface Design • Client-Server Architecture • Agile Methodologies
**Soft Skills**
Problem-solving • Team collaboration • Clear communication • Time management

## Projects

**Fandom TUI** | Java, Lanterna TUI Framework, OkHttp HTTP Client, Jackson JSON Parser, Maven

- Built a Java-based terminal application for browsing Fandom wikis, featuring article search and readable content display through a fully keyboard-navigable text interface.
- Integrated the MediaWiki API to fetch search results and parse wikitext content, implementing custom markup processing to render bold, italic, and structured text in the terminal.
- Designed a modular architecture with decoupled API client, content parser, and UI layers, demonstrating clean separation of concerns and maintainable code structure.
- Utilized Lanterna's terminal GUI components to create an intuitive multi-screen interface with scrollable views, search functionality, and contextual status indicators.

**Robot World Client-Server System** | Java, Sockets, Multithreading, JSON, TCP/IP

- Engineered multithreaded TCP server managing 20+ concurrent robot clients and thread-safe world state with obstacles.
- Designed JSON protocol for command/response messaging, enabling cross-compatible client-server communication.
- Built zero-knowledge client that learns world state solely through server responses.
- Applied synchronized concurrency controls to prevent race conditions during simultaneous robot updates.

## Experience

**Front Desk Assistant** (Volunteer) | **WeThinkCode_** | October 2025 – April 2026

- Managed concurrent inquiries and appointment scheduling, developing prioritization and task-switching skills
- Maintained accurate digital records and logs, ensuring data consistency across shifts
- Communicated clearly with visitors and staff, translating needs into actionable next steps
- Resolved scheduling conflicts and walk-in requests calmly under time pressure